CHARBEL DIB

PhD applicant | Machine learning, NLP & evaluation

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PROFILE

ENSTA engineering graduate with research experience in machine learning for additive manufacturing and RAG system evaluation. First author of an IEEE paper. Seeking a PhD in AI in France, with interests in learning from data, natural language processing and experimental model evaluation.

EDUCATION

ENSTA, Brest campus | Engineering degree (Master's level) | 2024 - 2026

Specialisation: Observation Systems and Artificial Intelligence.

Coursework: machine learning, deep learning, computer vision, signal processing and data fusion.

PUBLICATION

Charbel Dib, Rima Hleiss. *Cluster-Based Learning for Relative Density Prediction in Laser Powder Bed Fusion.* 2026 IEEE International Conference on Smart Sustainable Systems for Computer and Engineering Applications (3SCEA), 2026.

First author · DOI: 10.1109/3SCEA68071.2026.11603067

RESEARCH EXPERIENCE

Astek Innovation Lab | Final-year research internship in NLP / RAG

La Défense, France · Mar - Sep 2026

Research question: how can retrieval, evidence-extraction and classification errors be distinguished in a climate fact-checking system?

- Established a shared experimental framework to compare pipeline components: chunking, dense representations, hybrid retrieval and reranking.
- Studied LLM evidence extraction and prompt variants; evaluated a Phi-3 classifier previously adapted with LoRA, separating the contribution of each stage.
- Compared retrieval and classification metrics (NDCG, recall, macro-F1) and investigated LLM-as-a-judge assessment of evidence coverage.
- Analysed error propagation across the pipeline and documented evaluation limitations; organised runs with Docker, caching and experiment logs.

CESI / LINEACT | Machine learning research internship

Nanterre, France · May - Aug 2025

Research question: how do laser powder bed fusion (L-PBF) process parameters relate to the relative density of manufactured parts?

- Built and structured an experimental dataset linking manufacturing parameters to part properties.
- Compared predictive models and investigated clustering of manufacturing configurations to analyse relationships between process settings and quality.
- Contributed to formalising the research in a first-author IEEE conference paper with Rima Hleiss.

SCIENTIFIC & TECHNICAL PROJECTS

ENSTA document assistant | Academic project · 2026

Supervisor: Dr Abdelmalek Toumi · ENSTA

- Designed a bilingual RAG prototype: web/PDF/XLSX collection and cleaning, multilingual-E5 embeddings, a FAISS index and a multi-session FastAPI service.
- Implemented citations, refusal below a similarity threshold and human escalation; automated daily rebuilding of the web corpus and its index.
- Documented functional tests and prototype limitations in a project report; large-scale evaluation and structured user validation remain future work.

FleetGuard | Personal project · 2026

- Investigated imbalanced APS failure classification on the public Scania dataset: baselines, duplicate-group data splits, calibration and cost-based threshold selection.
- Froze preprocessing, model and threshold before final evaluation. Official test set of 16,000 observations: 95.73% recall, 43.05% precision, 16 false negatives and 475 false positives.
- Published a model card documenting the protocol, results and limitations; exposed scores through a FastAPI service and React interface. Public-data study without real-fleet validation.

ServicePilot | Personal project · 2026

- Designed a support workflow using LangGraph, FastAPI and SQLite: procedure and incident retrieval, a frozen proposal, human approval and server-side write authorisation.
- Developed an SDK, tests and offline evaluation scenarios on synthetic data; deterministic default mode with an optional LLM adapter.

METHODS & TOOLS

Evaluation: baseline comparisons, error analysis, NDCG, precision/recall/F1, calibration, threshold selection, documentation of protocols and limitations.

ML / NLP: Python, PyTorch, scikit-learn, XGBoost, Transformers, embeddings, hybrid retrieval, reranking, RAG; Pandas, NumPy, SQL.

Prototyping: FAISS, FastAPI, LangGraph, Docker, Git, Linux.

AWARD & LANGUAGES

1st prize, SPASEA Hackathon 2025 (Naval Group / DGA / ENSTA). Team contribution to **posAIdon**, a maritime copilot combining AIS, radio-frequency and ephemeris data for anomaly analysis.

Languages: French fluent · English fluent · Arabic native.